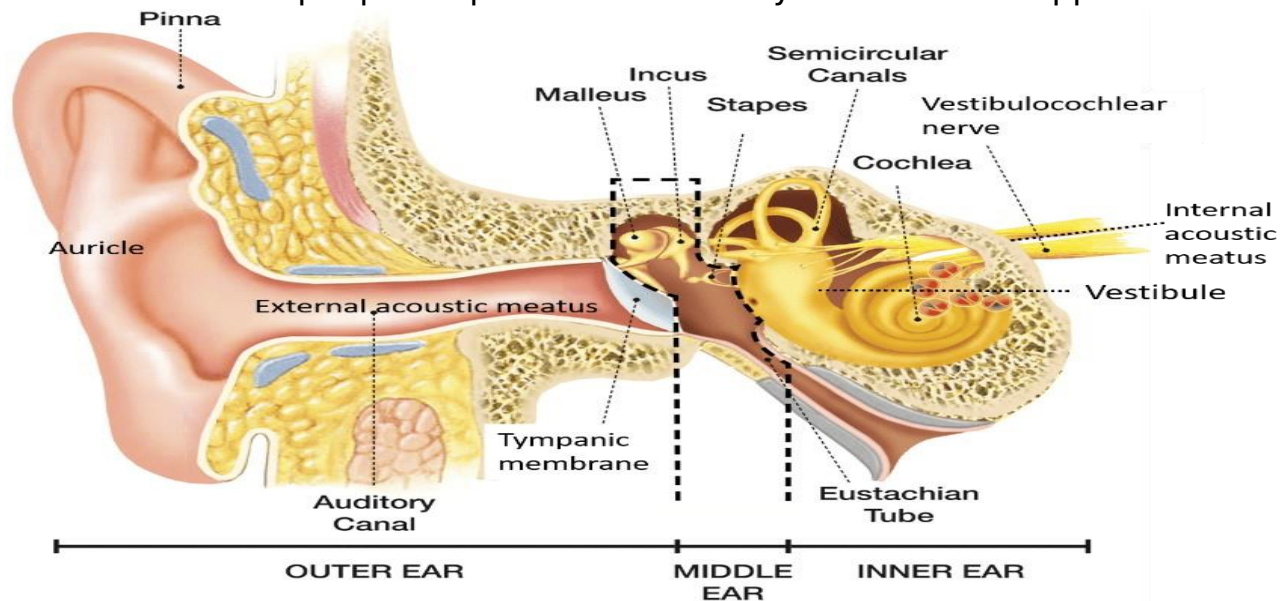


- Regions in brain have **no BBB (circumventricular organs)**: they include **posterior pituitary, lamina terminalis, median eminence of hypothalamus, pineal gland and area postrema.**

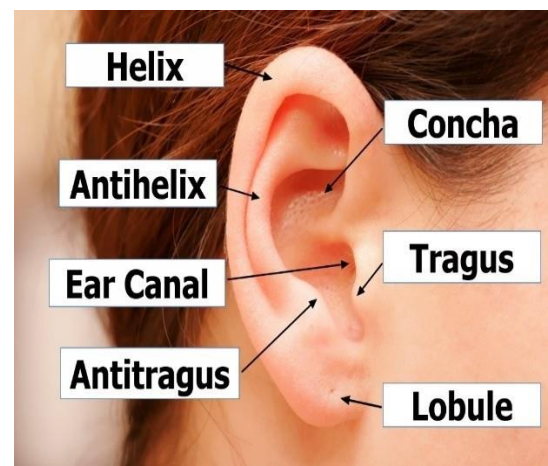
Lecture (10): Structure of the ear

- The ear is divided into external, middle and internal ear.
- It houses the peripheral parts of the auditory and vestibular apparatus.



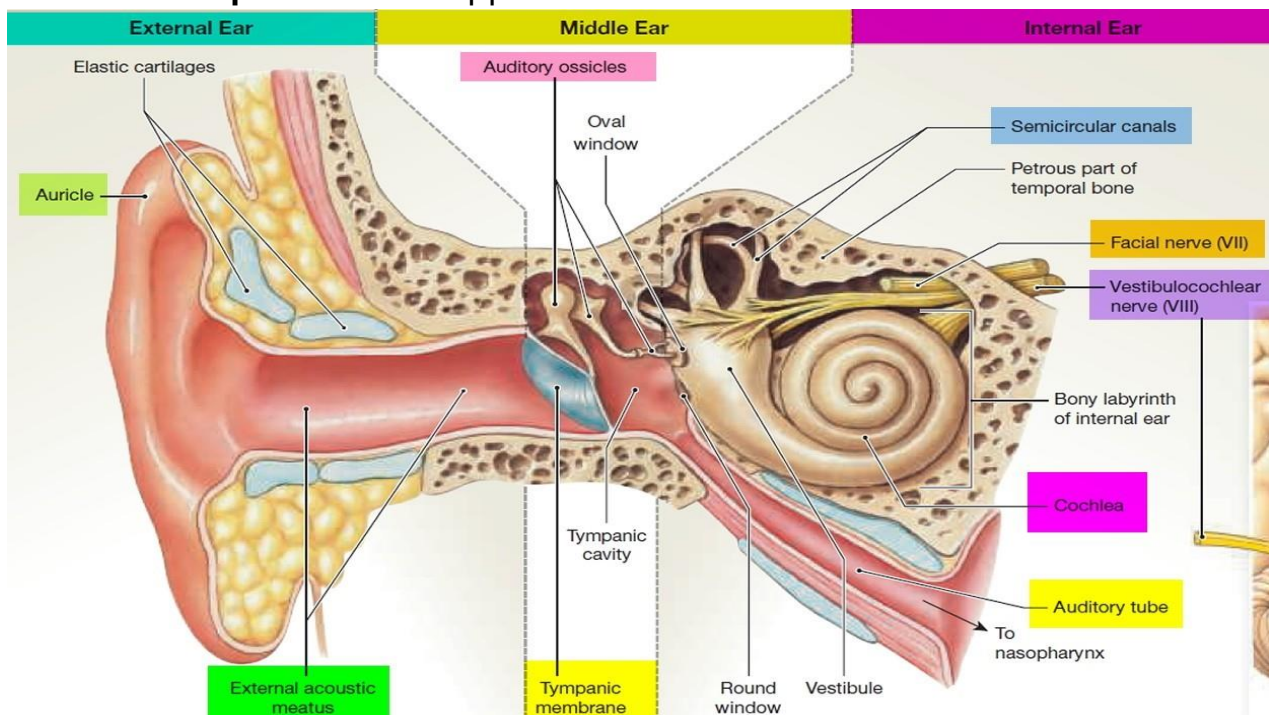
External Ear

- Consists of **auricle (pinna) & external acoustic meatus.**
- Collects and directs sound waves.
- The auricle:
 - It is formed of **resilient yellow elastic cartilage** except the **lobule** of the ear which is **soft fibro-fatty tissue.**
 - Small anterior, superior and posterior auricular muscles attach the auricle to the scalp and skull. All are supplied by the **facial nerve.**
 - Sensory supply of the auricle:
 - 1- **Great auricular nerve** supplies the whole of the cranial surface of the auricle and the lower part of the lateral surface.
 - 2- **Auriculotemporal nerve** supplies the upper part of the lateral surface.



External acoustic meatus:

- It is a **sinuous tube** (S-shaped), nearly 3 cm in length.
- Its **outer third** is **cartilaginous**, its **inner two-thirds** is **bony**.
- Its bony part is formed tympanic part of temporal bone.
- Hairs, sebaceous glands, ceruminous glands are abundant in the skin of the cartilaginous part.
- Skin with **neither** ceruminous gland **nor** hairs in bony part.
- **The bottom** of meatus is formed by tympanic membrane.
- **Sensory supply:**
 - **Auricular branch of vagus nerve** supplies floor & posterior wall of the meatus.
 - **Auriculotemporal nerve** supplies roof & anterior wall of the meatus.



Tympanic membrane (ear drum):

- The tympanic membrane separates between the external acoustic meatus & the middle ear cavity.
- It is a thin fibrous structure covered externally with stratified squamous epithelium and internally with low columnar epithelium.
- The membrane is nearly **circular**, 1 cm in diameter, and lies obliquely at 55° with the external acoustic meatus, facing downwards, forwards and laterally.
- The **outer surface** of the membrane is **concave** while its inner surface is convex its maximum convexity inward called umbo.
- The **handle of the malleus** is firmly attached to the inner surface of the membrane.

- **Parts of the membrane:**

1- **Pars flaccida:** upper triangular part of the membrane is **lax**.

2- **Pars tensa:** The main part of the membrane is **tense**.

- **Nerve supply:**

1- **Outer surface:** Auricular branch of **vagus nerve**.

2- **Inner surface:** Tympanic plexus of **glossopharyngeal nerve**.

- **Blood supply:** deep auricular artery (maxillary artery) & stylomastoid artery (posterior auricular).

- When the drum is illuminated for inspection, the concavity of the membrane produces a 'cone of light' radiating from the umbo over the anteroinferior quadrant.

Middle Ear

- The middle ear is an air space (biconcave lens-shaped) in the temporal bone.

- The tympanic cavity, about 15 mm in anteroposterior and vertical diameters.

- **Transmits and amplifies sound vibrations** from tympanic membrane laterally to inner ear medially through **auditory ossicles**.

- Nerve supply of middle ear: by **Tympanic plexus from glossopharyngeal nerve**. **Tympanic cavity boundaries:**

- **Lateral wall:** tympanic membrane & Epitympanic recess which is upward extension of of tympanic cavity.

- **Medial wall:** which is the lateral wall of inner ear and is formed of:

- **Promontory of the 1st turn of cochlea.**

- **Horizontal part of bony facial canal.**

- **Oval window, closed by foot plate of stapes.**

- **Round window, closed by secondary tympanic membrane.**

- **Roof: tegmen tympani** of petrous bone that separate tympanic cavity from middle cranial fossa.

- **Floor:** a thin plate of bone above the jugular fossa.

- **Posterior wall:**

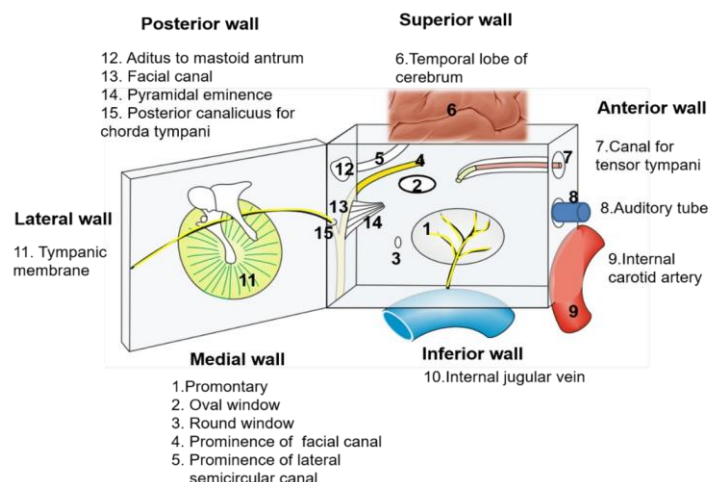
- Aditus of mastoid antrum.

- Vertical part of bony facial canal.

- Pyramid gives origin to stapedius muscle.

- **Anterior wall:** is perforated by 2 openings:

- Upper opening for canal of



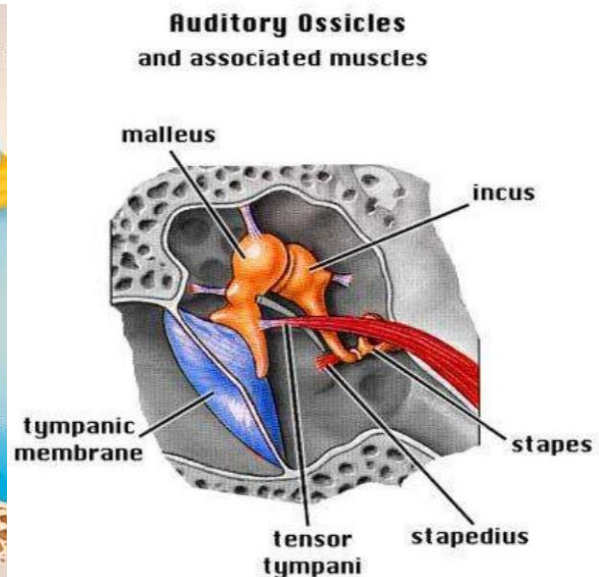
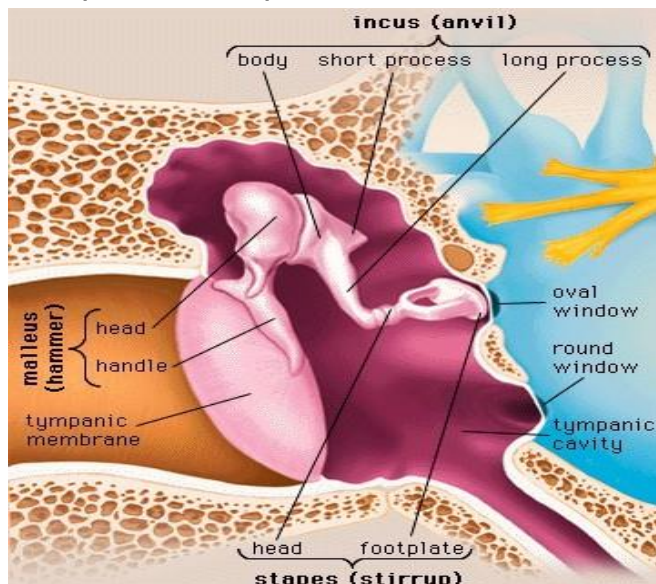
tensor tympani muscle.

- lower opening for the auditory tube.
- Its lower part is related to carotid canal.

Contents of middle ear cavity:

Three auditory bony
ossicles:

- They are from lateral to medial: **Malleus, Incus & Stapes**. They articulate by synovial joints.
- Handle of Malleus is fixed to the inner side of tympanic membrane, while the foot plate of Stapes covers the oval window.



• Two muscles:

- **Tensor tympani:** arises from bony part of the auditory tube and inserted into handle of Malleus. It is supplied by mandibular nerve.
- **Stapedius:** arises from the pyramid and is inserted into the stapes. It is supplied by facial nerve.
- **Action of both muscles** prevent over vibrations of tympanic membrane by loud sounds to protect inner ear.

■ Nerves:

- **Chorda tympani:** branch of facial nerve that lies over the inner surface of tympanic membrane.
- **Tympanic plexus:** from glossopharyngeal nerve. It lies over the promontory.

Auditory tube (pharyngotympanic tube, Eustachian tube):

- The auditory tube connects the **nasopharynx with the middle ear cavity**.
- About 3 cm long, it slopes from the middle ear forwards, medially and downwards.
- It has bony and cartilaginous parts.
- **Lateral bony part is 1 cm long while medial cartilaginous part is 2 cm long.**
- During swallowing, the tube is opened to allow equalization of air pressure on the two sides of the tympanic membrane.
- The nerve supply is by pharyngeal branch of the **pterygopalatine ganglion** (maxillary nerve) and the tympanic plexus (**glossopharyngeal nerve**).

Internal Ear

- The internal ear consists of:
 - 1- **Bony (osseous) labyrinth:** a complex series of connected cavities inside the petrous temporal bone.
 - 2- **Membranous labyrinth:** complex series of connected cavities inside bony labyrinth.
- The fluid inside the membranous labyrinth called **endolymph** while the fluid between bony and membranous labyrinth called **perilymph**.
- Internal ear is concerned with **hearing and equilibrium**.

Parts of bony labyrinth: in order from front to back:

1- Cochlea:

- **Cochlea** is a conical snail-shaped cavity in petrous bone.
- It consists of two and **three-quarter** spiral turns.
- It makes its spirals around a central bony core known as the **modiolus**; the base of which rests at the bottom of the internal acoustic meatus, while the apex projects laterally (towards the middle ear).
- The 1st turn of the cochlea at its base is the largest turn & projects by **promontory** into the cavity of middle ear.

2- The Vestibule:

- **Vestibule** is a hollow in petrous bone, located between the cochlea and the semi-circular canals.
- It contains the **membranous saccule and utricle**.
- The scala vestibuli of the cochlea opens into the front of the vestibule and the **orifices** of the semicircular canals open in the back of it.
- The medial wall abuts on the internal acoustic meatus, is perforated by branches of vestibular nerve.

- The **oval window** lies on the lateral wall of the vestibule and is closed by the foot-plate of the stapes.

3- The 3 Semicircular Canals:

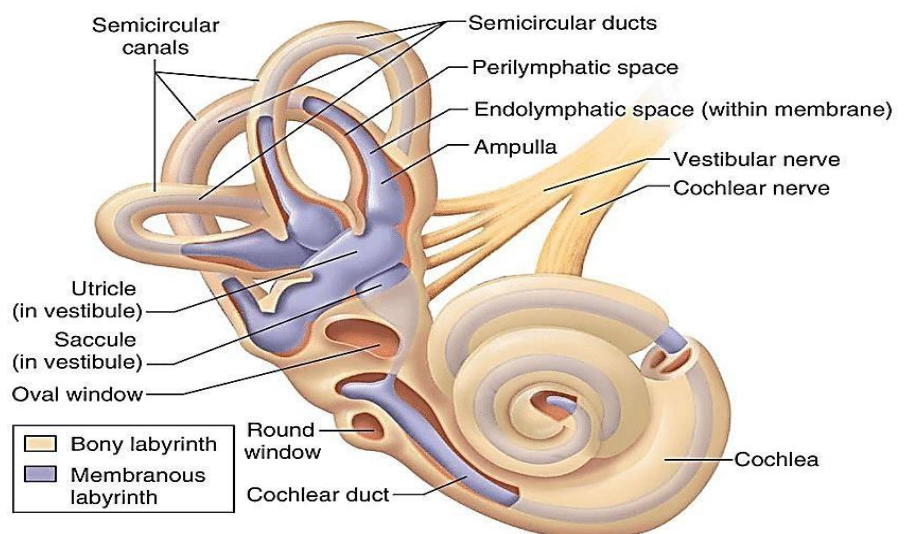
- **Superior, posterior and lateral semicircular canals** project from the back of the vestibule, and they lie in three planes at right angles to each other.
- At one end of each canal is a swelling called the bony ampulla. The ampullae of all three canals open into the vestibule.
- Each canal is about two-thirds of a circle.

■ Membranous Labyrinth:

- It consists of one continuous closed cavity containing endolymph.
- It lies within the bony labyrinth and is surrounded by perilymph.
- It is composed of the **cochlear duct, saccule, utricle & 3 semi-circular ducts. The cochlear duct:**
 - The cochlear duct is situated within the cochlea. It is the anterior part of the membranous labyrinth.
 - It contains sound receptors (spiral organ of Corti).
 - It is triangular in cross-section. It is surrounded by the basilar membrane which supports the spiral organ of Corti, and vestibular membrane which passes obliquely across the cochlea.
 - The presence of the cochlear duct creates two canals; above it (scala vestibuli) and below it (scala tympani).
 - The scala tympani communicates with the scala vestibuli at the apex of the modiolus.
 - It communicates with middle ear cavity via the round window, which is closed by secondary tympanic membrane.

Utricle & Saccule:

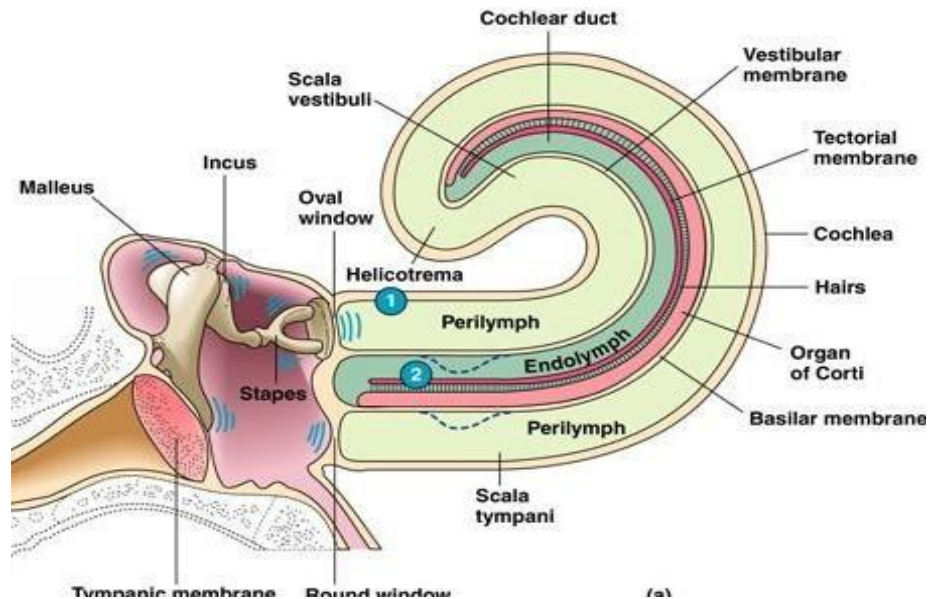
- The **utricle & saccule** are two membranous sacs located in the vestibule. The utricle is the larger of the two.
- The saccule receives the cochlear duct and utricle receives the three semi-circular ducts.
- They contain sensory



receptors called **maculae** of the utricle and saccule which detect **linear acceleration** of the head in the vertical and horizontal planes.

Three semi-circular ducts:

- They are located within the semi-circular canals.
- They contain sensory receptors (**3 cristae ampullaris**) in the ampullae of the semi-circular canals for kinetic balance of head movement (sensitive to angular acceleration of the head).



Blood supply of the labyrinth:

- **Labyrinthine artery** from the basilar artery.
- **Stylomastoid artery** from posterior auricular artery.

Lecture (11): Spinal cord I

Gross anatomy of the Spinal Cord:

• Position:

The spinal cord connects the brain to the peripheral nervous system (PNS). It is the caudal part of the central nervous system that occupies the upper two-thirds of the vertebral canal.

• Shape:

The spinal cord is roughly cylindrical in shape, 45cm long.

• Beginning:

It begins superiorly at the foramen magnum in the skull base, where it is continuous with the medulla

